

A Dual-View Fusion Network for Automatic Spinal Keypoint Detection in Biplane X-ray Images

Dandan Zhou¹, Lijun Guo^{*12}, Rong Zhang¹², Xiuchao He³, Qiang Wang³, Jianhua Wang³

¹ The Faculty of Electrical Engineering and Computer Science, Ningbo University, Ningbo, China

² The Zhejiang Key Laboratory of Mobile Network Application Technology, Ningbo, China

³ The Department of Radiology, The Affiliated Hospital of Medicine School of Ningbo University, Ningbo, China

* Corresponding author, e-mail: guolijun@nbu.edu.cn

Abstract—Accurate keypoint detection in medical images of the spine is critical for the assessment, diagnosis, treatment planning, and clinical investigation of spinal deformities. However, due to severe occlusions of spinal structures in lateral X-ray images, accurate keypoint detection can be hardly achieved in lateral X-ray images based on single-view information. Thus, methods based on both the anterior-posterior (AP) and lateral (LAT) X-ray image views have been proposed to alleviate occlusion problems and achieve better keypoint detection performance. Although some progress has been made with these dual-view methods, they do not effectively exploit a priori knowledge of the spine and hence cannot adequately account for the structural correlation of the vertebrae across views. In this paper, a new dual-view fusion network (DVFNet) framework is proposed for keypoint detection in spinal X-ray images. This framework obtains structural correlations between AP and LAT views of the spine based on a priori spine knowledge represented by high-level semantic features. Meanwhile, the proposed framework combines local and global features extracted respectively by a local subnetwork and a global subnetwork. On the one hand, the local subnetwork is constructed as an enhanced codec structure based on both the AP and LAT views. This subnetwork is trained to output local features that contain both joint semantic features of the two views and independent fine-grained features of each individual view. This scheme leads to accurate keypoint estimation locally. On the other hand, the global subnetwork utilizes a self-attention mechanism to extract view-specific global features based on either the AP view or the LAT view in order to eliminate ambiguity, and reduce confusion on keypoint locations. Further, we propose a weighted feature fusion (WFF) module for adaptive fusion of the local and global features. We evaluated the DVFNet model on a private dataset and found that our proposed method achieves more accurate spinal keypoint detection compared to other state-of-the-art methods, and thus our method can provide reliable assistance to clinicians.

Index Terms—spinal keypoint detection, biplane X-ray images, dual-view feature fusion

I. INTRODUCTION

Spinal deformities [1] are common spinal disorders that are manifested by abnormal curvatures of the human bony vertebral column. During clinical diagnosis of spinal deformities, morphological parameters (such as the Cobb angle, the lumbar lordosis angle, and the thoracic kyphosis angle) are typically measured based on the locations of the spinal keypoints in the anterior-posterior (AP) and lateral (LAT) X-ray images [2]. Therefore, accurate keypoint detection is a crucial step in computer-aided diagnosis of spinal deformities. However, manual identification of the four corner points of each of the

seventeen thoracic and lumbar vertebrae is time-consuming. Also, the accuracy of labeling these points depend heavily on several factors including the experience of the human annotator, subject-specific variations in vertebral structures, and occlusion of vertebrae by soft tissues of the thoracic cavity. Therefore, effective automation of the spinal keypoint detection process is essential for eliminating the subjective labeling errors and boosting the detection performance.

In recent years, several methods for automatic keypoint detection have been investigated, among which great potential has been especially demonstrated by deep learning methods [3]. According to the type of the network prediction target, these latter methods can be categorized into coordinate regression methods and heatmap methods. The coordinate regression methods usually employ fully connected layer to directly predict the coordinates of the keypoints [4] [5]. Such methods typically show strong robustness and utilize global features to better maintain reasonable overall shapes of the predicted keypoint locations. However, since the weights obtained by these methods heavily depend on the distribution of the training data and thus lack the ability for spatial generalization, these methods cannot accurately predict all keypoint coordinates. The heatmap methods [12] [13] focus more on utilizing local information as convolution kernels are applied to feature maps. Therefore, these methods are good at precisely locating individual keypoints within local regions. However, under occlusion and blur conditions, these methods perform poorly in terms of keypoint continuity and stability [6]. By contrast, transformer-based methods [9] have a natural advantage in capturing global dependencies. Indeed, these methods are inspired by recent progress in natural language processing (NLP), where convolutional neural network (CNN) architectures are combined with self-attention mechanisms [10] in order to effectively capture the dependencies of individual keypoints in image data and hence achieve enhanced performance on keypoint detection tasks.

Most of the recent methods for spinal keypoint detection focus on AP X-ray images and have achieved remarkable results. However, in LAT X-ray images of the spine, the vertebrae are often heavily occluded by soft tissues and organs within the thoracic cavity (as shown in the blue box in Fig.1), and this occlusion phenomenon undermines the accuracy of automatic keypoint localization based only on the single-view

LAT X-ray images. To address this problem, several methods based on dual-views of the spine have emerged [14] [15] [16] in recent years. In these methods, both the AP and LAT X-ray images of the spine of a patient are utilized to obtain more comprehensive information, and achieve better detection performance. However, these methods usually use only one of the coordinate regression or heatmap approaches for keypoint detection, and thus these methods cannot effectively use local and global features simultaneously. On the one hand, the local features provide precise information for keypoint localization. However, the high similarity between vertebrae leads to similar local feature representations, and this poses an additional challenge of determining which vertebrae each keypoint belongs to. Therefore, keypoint prediction based on local features might lack robustness and do not usually maintain a good overall spine shape. On the other hand, the global features take into account contextual information of the overall spine structure. These features help with keypoint localization and removal of the keypoint ambiguity arising from local feature similarity. However, the global features tend to overlook some details and local variations, and thus usually lead to low-accuracy keypoint localization [7] [8]. Therefore, the existing dual-view methods, which utilize either global features or local features (but not both) suffer from either low accuracy or poor robustness. In addition, for dual-view feature fusion, most of these dual-view methods employ shallow fine-grained features (such as color and texture features), which cannot effectively capture the correlation between the AP and LAT views of the spine. In fact, the correlation between these views is usually established through shared anatomical structures and spatial relationships. Therefore, from the perspective of the a priori knowledge of the spine, the correlation between the two views can be captured more comprehensively and accurately through the utilization of rich deep semantic features. These features help overcome the problems of occlusion and ambiguity and lead to better dual-view-assisted keypoint detection performance. In this paper, we propose a new dual-view

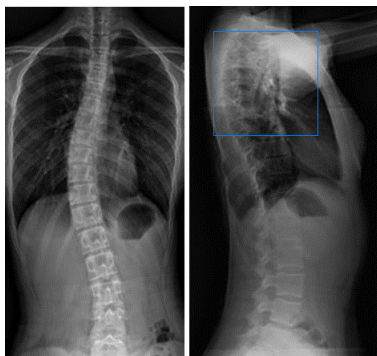


Fig. 1. An example of the AP and LAT X-ray images of a patient. Keypoint detection in LAT view is unreliable due to vertebral occlusion by the ribcage as indicated by the blue box.

fusion network (abbreviated DVFNet) framework, in which the correlation between the AP and LAT X-ray views of the spine is exploited for efficient and accurate automatic keypoint

detection. Specifically, DVFNet consists of two parts: a local subnetwork and a global subnetwork. The local subnetwork is constructed based on an enhanced U-Net encoder-decoder architecture which effectively utilizes the a priori knowledge of the spine to capture the correlation between the AP and LAT views through high-level semantic features. Specifically, high-level semantic features of the two views are generated by the encoder in the local subnetwork. Then, feature fusion is applied such that the connection between the multi-view spatial morphological features of the spine can be effectively captured. In addition, the local features output by the decoder includes joint semantic features of the two views as well as view-specific fine-grained features. These features facilitate keypoint estimation locally. The global subnetwork is constructed based on each of the AP and LAT views separately, where the global information of each view is modeled via a self-attention mechanism which captures the global contextual information and eliminates the keypoint location ambiguity. The local and global features are thus adaptively fused by a weighted feature fusion (WFF) module in order to improve the accuracy and robustness of the keypoint detection process. Also, a deep-supervision loss is used for network parameter optimization. In summary, the main contributions of this paper are as follows.

- We proposed DVFNet, a new dual-view fusion network framework for spinal keypoint detection. The framework exploits a priori knowledge of the spine and uses deep semantic features to effectively capture the correlation between the AP and LAT X-ray views of the spine, and thus boost the keypoint detection performance.
- DVFNet is based on a new subnetwork architecture that combines local and global features of the AP and LAT X-ray views. In the local subnetwork, dual-view high-level semantic features are fused, while view-specific features are retained. This leads to accurate local estimates of keypoint locations. In the global subnetwork, the global information of each image view is modeled using a self-attention mechanism in order to enforce global shape constraints, and reduce the keypoint location ambiguity. The keypoint detection accuracy and robustness are thus further improved through weighted fusion of local and global features.
- We constructed a dataset of 326 cases with AP and LAT X-ray images for the whole spine in each case. This dataset is intended to promote research on spinal deformity studies. The experimental results show the superiority of our method compared with other state-of-the-art methods.

II. METHODS

A. Dual-View Fusion Network (DVFNet) Framework

As shown in Fig.2, the overall DVFNet framework consists of four parts: (1) The local subnetwork, which uses a full convolutional structure to simultaneously process both the AP and LAT views of the spine based on a local perception

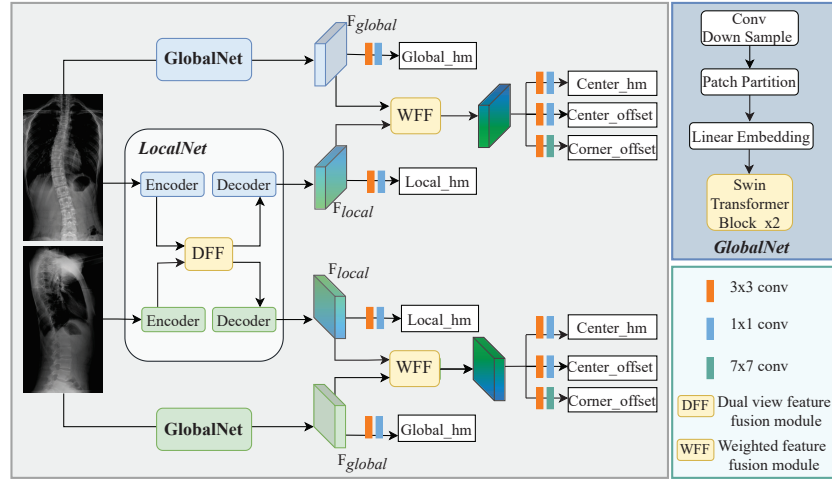


Fig. 2. The proposed DVFNet framework.

mechanism, and hence generate local joint features of the two views and view-specific features. (2) The global subnetwork, which learns remote dependencies between keypoints based on a self-attention mechanism in order to generate global contextual features. (3) The weighted feature fusion (WFF) module, which adaptively fuses local and global features to generate robust feature representations for accurate spine keypoint detection. (4) The overall loss function for network parameter optimization, where the overall loss is a weighted sum of the loss of each prediction target and the deep-supervision loss.

B. The Local Subnetwork

As shown in Fig.3, we constructed the local subnetwork with two branches for the AP and LAT views and encoding and decoding stages for each branch. The encoder backbone is a residual network (ResNet) which possesses good learning capabilities. In this subnetwork, the AP and LAT X-ray images are fed as inputs to the two respective AP and LAT image encoders. Then, feature maps are extracted at the outputs of the last residual blocks for the 2nd, 3rd, 4th, and 5th convolutional layers of each encoder. So, four feature layers of different scales are obtained for each of the AP and LAT image encoders, and these layers are denoted as A_{Ei} and L_{Ei} , $i = 1, 2, 3, 4$, respectively. Moreover, the a priori knowledge of the spine shows a semantic connection between the AP and LAT views of the spine. Therefore, we design the dual-view feature fusion (DFF) module at the end of the two encoders to capture the correlation of the two views, and hence improve the discriminability of the vertebral shapes and keypoint features. The architecture of the DFF module is shown in Fig.3. Firstly, the high-level semantic features A_{E4} and L_{E4} of the AP and LAT branches are simultaneously fed into the DFF module. The AP and LAT features are thus spatially summed up. Then, the obtained sums are respectively spliced with the original view-specific features A_{E4} and L_{E4} in the channel direction. Finally, the channel size is adjusted

through 1×1 convolution. The A_{D4} and L_{D4} features output from the DFF module are fed into the AP and LAT decoders, respectively. This completes the combination of the low-level detailed information and the high-level semantic information. Finally, the local features A_{D1} and L_{D1} (corresponding to F_{local} in Fig.2) extracted from the AP and LAT branches of the local subnetwork contain joint semantic features of the two views as well as view-specific fine-grained features. The structural design of the local subnetwork can effectively overcome the occlusion problem of the LAT view, and thus the keypoint detection accuracy can be significantly enhanced in local regions.

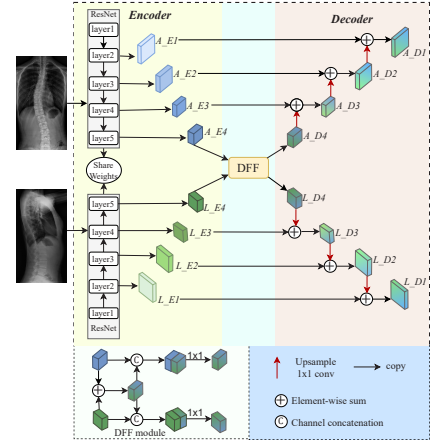


Fig. 3. The framework for the local subnetwork and its components.

C. The Global Subnetwork

The human spine presents a stable topology. So, the relative positional relationships between keypoints of different vertebrae can be used as effective constraints for keypoint detection. Therefore, we designed the global subnetwork with branches for AP and LAT X-ray images of the spine in order to further enhance the global discriminative power of

the proposed DVNet and reduce the keypoint localization ambiguity. The structure of the global subnetwork is shown in Fig.2, and we use a self-attention mechanism to model global dependencies and achieve high efficiency with the Swin Transformer [11] framework. This modeling approach provides global shape constraints, and thus eliminates the ambiguity of keypoint locations.

D. The Weighted Feature Fusion Module

Due to the shallow network structure used in constructing the global subnetwork, the global features may be contaminated by image noise. Thus, simply fusing local and global features may introduce image noise, and hence undermine the keypoint detection performance. Therefore, we propose an adaptive method for fusing local and global features. Specifically, the weighted feature fusion (WFF) module preprocesses the global features through linear and nonlinear transformation layers to highlight the most discriminative features and weaken the background and noise effects. This module can be mathematically expressed as

$$F_{fusion} = \Phi [F_{local}, \tanh(F_{global} \bullet W + B)] \quad (1)$$

where F_{global} and F_{local} respectively denote vectors of global and local features, while W and B are respectively weight and bias vectors with the same dimension as F_{global} . The dot ($\bullet$) denotes the elementwise dot-product operator, $\tanh$ is the hyperbolic tangent function, and Φ denotes a concatenation function and a 1×1 convolution operation. The WFF module tends to fuse the global and local features more effectively and improve the network robustness.

E. The Loss Function

As mentioned above, the proposed DVNet architecture is a network with two branches for the AP and LAT X-ray images, where each branch performs keypoint detection based on single-view global features as well as local features obtained via dual-view fusion feature, with three detection objectives (namely, a center heatmap, a center offset and a corner offset). These three detection objectives together determine the keypoint locations and their losses constitute the main network loss. The deep-supervision losses of the local and global subnetworks for the center heatmap account for a smaller proportion of the total network losses.

1) *Ground-truth keypoint locations*: We used earlier keypoint estimation strategy [14] to first localize each vertebra by the vertebral center heatmap and then employ the corner offset to calculate the corner-point coordinates. The center offset is used to eliminate the quantization error of the vertebral center.

Center heatmap: As shown in Fig.4 (a) and (b), each X-ray image in our dataset is labeled with 4 corner-point coordinates of 17 vertebrae (12 thoracic and 5 lumbar vertebrae). The center k of each vertebra was found from the corner-point coordinates and an un-normalized Gaussian kernel was used to generate a ground-truth single-channel center heatmap as shown in parts (c) and (d) of Fig.4. The center heatmaps were all optimized based on the focal loss [17].

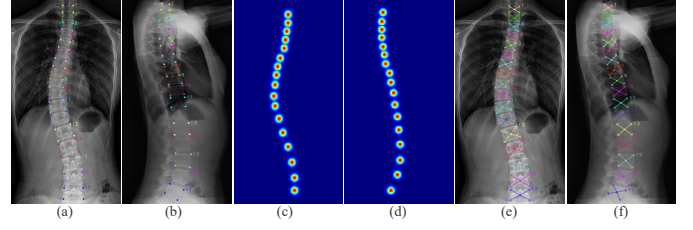


Fig. 4. The AP and LAT keypoints, center point heatmaps, and corner-point offsets:(a)(b) Each X-ray image shows 17 vertebrae, and each vertebra has 4 corner landmarks, for a total of 68 landmarks per image.(c)(d) Vertebra center heatmap. (e)(f) Corner offset.

Center offset: The center offset is used to reduce the effect of the quantization error caused by down-sampling. Specifically, the location (x, y) of the center of a vertebra on an input image is mapped to a feature map that is down-sampled n times and is denoted as $(\lfloor \frac{x}{n} \rfloor, \lfloor \frac{y}{n} \rfloor)$. The center offset is defined as $(\frac{x}{n} - \lfloor \frac{x}{n} \rfloor, \frac{y}{n} - \lfloor \frac{y}{n} \rfloor)$. Training is performed with the L1 loss.

Corner offset: As shown in parts (e) and (f) of Fig.4, the corner offset is expressed as a vector pointing to the 4 corner points of the vertebral body from the center point. This offset is denoted as $(\frac{x}{n} - \frac{x_c}{n}, \frac{y}{n} - \frac{y_c}{n})$, where $(\frac{x_c}{n}, \frac{y_c}{n})$ are the coordinates of the corner points of the vertebrae. Training is performed with the L1 loss.

2) *Loss computation*: In order to train the proposed dual-view network in an end-to-end manner, an overall loss function is defined as follows:

$$Loss = L_{AP} + L_{LAT} \quad (2)$$

where L_{AP} and L_{LAT} represent the losses of the AP and LAT branches of the spine, respectively. The sum of the losses of the two branches, Loss, is used to train the model and optimize the parameters. Here, L_{AP} and L_{LAT} are, respectively, defined as:

$$L_{AP} = \alpha L_{local_hm} + \beta L_{global_hm} + \gamma L_{center} + L_{reg} + L_{wh} \quad (3)$$

$$L_{LAT} = \alpha L_{local_hm} + \beta L_{global_hm} + \gamma L_{center} + L_{reg} + L_{wh} \quad (4)$$

where L_{local_hm} and L_{global_hm} represent the deep-supervision losses of the center heatmap for the local and global subnetworks, respectively. The regularization parameters are set as $\alpha = \beta = 0.2$ and $\gamma = 0.6$. The losses L_{center} , L_{reg} and L_{wh} are computed by the network based on the global and local features for the center heatmaps, center offsets, and corner offsets, respectively.

III. EXPERIMENTS AND RESULTS

A. Dataset Description

The lack of datasets of adequate sizes is a major obstacle for research in medical image analysis. It seems that this problem will remain unavoidable because of the difficulty of data acquisition, high annotation costs, and issues of patient privacy. To the best of our knowledge, there is no publicly

available dataset of LAT X-ray images of the spine. For this study, we collected whole-spine AP and LAT X-ray images for 328 patients at local hospitals. The images were annotated by two experienced spine surgeons. As shown in Fig.4 (a) and (b), each image contains a total of 17 vertebrae from the first thoracic vertebra (T1) to the fifth lumbar vertebra (L5). Each vertebra is labeled with 4 corner points for a total of 68 keypoints in each image. In this paper, the whole dataset is subdivided with 80% for training and 20% for testing.

B. Experimental Settings

Our DVFNet framework was implemented in PyTorch and run on a computer with NVIDIA GeForce RTX 3090 Ti (with a 24 GB of memory). To reduce the chances of overfitting, the collected dataset was augmented with additional samples through random flipping, random scaling, and contrast variation. Each original image was cropped to 1024×512 pixels and then used as a network input. In addition, an Adam optimizer was used, the batch size was set to 4, the initial learning rate was 1e-5, and training was carried out for a total of 100 epochs.

C. Evaluation

In this paper, we evaluated the keypoint detection performance using the average detection error (ϵ_{dec}) and the normalized average absolute error (ϵ_{MAE}). These two evaluation metrics are defined as:

$$\epsilon_{dec} = \frac{1}{M} \sum_{k=1}^M \sqrt{(gt_k^l - pre_k^l)^2} \quad (5)$$

$$\epsilon_{MAE} = \frac{1}{M} \sum_{k=1}^M (|gt_k^l - pre_k^l|) \quad (6)$$

where M is the number of keypoints per image, and gt_k^l and pre_k^l are respectively the ground-truth and predicted keypoint coordinates (these coordinates must be normalized first for ϵ_{MAE} calculation). Both evaluation metrics are required to be as low as possible for the best performance.

D. Comparison with Other Methods

To verify the effectiveness of the proposed method, we compared its performance against state-of-the-art methods on a private dataset. As shown in Table I, these reference methods are the single-view methods of Ao et al. [12] and Yi et al. [14], and the dual-view method of Zhang et al. [16]. In comparison with the abovementioned reference methods, the results show that our method reduces the ϵ_{dec} metric by 1.1134 and the ϵ_{MAE} metric by 0.00056 on the AP X-ray images, and also reduces the ϵ_{dec} metric by 1.83571 and the ϵ_{MAE} metric by 0.00046 on the LAT X-ray images. Not only does our method outperform the best single-view method, but also improves the results substantially compared to the dual-view method. Fig.5 shows qualitative prediction results for the compared methods on AP and LAT X-ray images of the same patient. The red and green dots represent ground-truth and predicted coordinates, respectively, while each blue line connects ground-truth and

TABLE I
COMPARATIVE EXPERIMENTAL RESULTS FOR THE PROPOSED METHOD AND THREE REFERENCE METHODS ON AP AND LAT X-RAY IMAGES.

Methods	Scaled ϵ_{MAE}		Scaled ϵ_{dec}	
	AP X-rays	LAT X-rays	AP X-rays	LAT X-rays
Ao et al.	0.00436	0.00707	5.74300	10.08641
Yi et al.	0.00596	0.00632	8.13231	8.62540
Zhang et al.	0.00827	0.01583	10.99091	22.63517
DVFNet	0.00380	0.00586	4.62870	6.78969

predicted coordinates, with a longer line representing a larger deviation of the keypoint prediction. The method of Ao et al. [12] can maintain the overall spine shape relatively accurately by introducing constraints on the global features. However, due to occlusions, the vertebrae can be easily mismatched with the keypoints in the LAT X-ray images as shown by the yellow box in the first column, and this leads to a low prediction accuracy. By contrast, the method of Yi et al. [14] avoids large prediction bias by performing keypoint detection based on local features. However, this method is susceptible to vertebral center and don't consider global shape constraints, and this leads to an overall shift of the vertebral corner points in the AP and LAT views (as shown in the yellow box in the second column). The method of Zhang et al. [16] is more susceptible to noise and background effects, and this undermines the network performance. This shortcoming is due to the fact that the a priori knowledge of the spine is not considered in the fusion of the two-view features. Therefore, the prediction results are highly biased, as shown in the yellow box in the third column. Compared to these three reference methods, our method fully utilizes the correlation between the two views by performing two-view feature fusion based on high-level semantic features on the one hand, and effectively fusing global and local information on the other hand. As a result, our method demonstrates better performance under occlusion conditions for LAT X-ray images. In this way, our method is able to provide more accurate and robust keypoint detection results.

E. Ablation Experiments

In order to verify the effectiveness of the proposed dual-view feature-fusion strategy based on high-level semantic information in detecting spinal keypoints, we conducted an ablation study where we observed the performance after adding the DFF module to each of the layers 2, 3, 4, and 5 of the ResNet-based encoder backbone, respectively. The results are shown in Table II. The best detection performance is achieved with layer-5 features (i.e., DVFNet in this paper). This verifies that our method better captures the structural correlation between the AP and LAT images. In addition, we conducted ablation experiments on the global subnetwork. As shown in Table III, adding global subnetworks to the AP and LAT branches on top of the local subnetwork further reduces the detection error for both views. This shows that

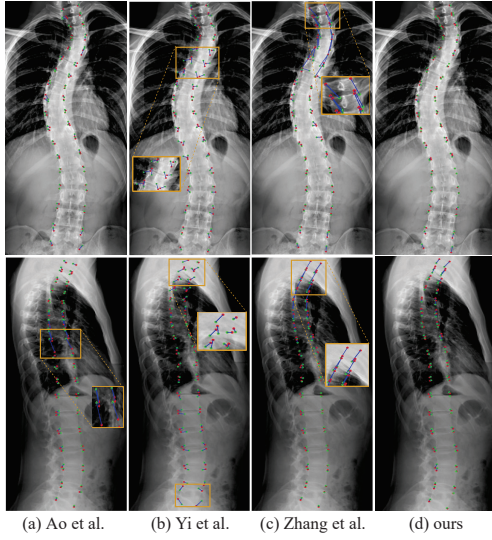


Fig. 5. Visualization of the keypoint detection results for different methods on a private dataset. The red dots represent ground-truth locations and the green dots represent the predicted locations.

the combination of the local and global features boosts the keypoint detection performance.

TABLE II
PERFORMANCE OF OUR METHOD WHEN THE DFF IS ADDED TO DIFFERENT LAYERS.

Methods	Scaled ϵ MAE		Scaled ϵ dec	
	AP X-rays	LAT X-rays	AP X-rays	LAT X-rays
layer2	0.00492	0.00672	6.73978	8.93285
layer3	0.00445	0.00592	5.70495	7.96926
layer4	0.00385	0.00594	4.69405	8.05049
layer5(DVFNet)	0.00380	0.00586	4.62870	6.78969

TABLE III
ABLATION EXPERIMENTS ON SOME GLOBAL SUBNETWORKS.

Methods	Scaled ϵ MAE		Scaled ϵ dec	
	AP X-rays	LAT X-rays	AP X-rays	LAT X-rays
localNet	0.00391	0.00590	4.76268	7.88823
localNet+globalNet (DVFNet)	0.00380	0.00586	4.62870	6.78969

IV. SUMMARY

This paper presents DVFNet, an end-to-end fully automated spine keypoint detection framework, which achieves highly accurate keypoint detection on both the AP and LAT views of the spine. First, DVFNet establishes structural correlation between the AP and LAT views of the spine using a priori knowledge of the spine, and thus mitigates the effects of image blurring and occlusion. Secondly, a new combined local and global subnetwork structure is proposed to improve the accuracy and robustness of keypoint detection through effective fusion of local and global features. Finally, a deep-supervision strategy is used for predicting the center heatmap,

which implicitly guides the network to approach the centroid coordinates more accurately. Our proposed DVFNet framework achieves accurate keypoint localization and can provide reliable assistance to clinicians.

ACKNOWLEDGMENT

This work was supported by the Zhejiang Provincial Public Welfare Technology Research Project(No. LGF21F020008), Ningbo Municipal Public Welfare Technology Research Project (No.2022S134).

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